INNOVERSE

FEASIBILITY AND VIABILITY

SMART INDIA HACKATHON 2026

Feasibility: Can we build it?

4 / 4 PILLARS READY

Technical

Own 2D shallow-water + SPH solvers in C# / .NET 8. Exports Delft3D FM and DualSPHysics cases.

- ✓ SPH near-field 1–3 km, 2D model 20–100 km
- ✓ Checked vs Ritter exact solution: L1 2.9%

Data

All inputs free: SRTM terrain tiles, Sentinel-1/2, OpenStreetMap, CWC / India-WRIS dam records.

- ✓ Takes lake detections from Sentinel-1/2 scans (GEE)
- ✓ Zero ground sensors needed

Financial

Open-source stack running on free cloud tiers. Runs on CPU, no GPU needed.

- ✓ ₹0 data and software licence cost
- ✓ No paid solver or GIS licence

Operational

Docker + job queue; offline mode with cached terrain for air-gapped (NTRO) use.

- ✓ One-command Docker deploy
- ✓ Outputs .shp, .kml, PDF, CAP

VALIDATION LADDER

1 Ritter / Stoker	2 Malpasset 1959	3 Teesta 2023 vs S-1
L1 2.9% / 0.7%	runner ready	CSI pending

⚠ DRAWBACKS (honest view, with how we handle each)

30 m terrain by default	Not truly real-time	Uncertain breach inputs	SPH vs 2D model differ
Coarse terrain limits fine local detail; finer DEMs (GLO-30, FABDEM) can be uploaded	Sentinel-1 revisit is 6–12 days; risk atlas pre-computes every dam to fill the gap	Froehlich is empirical; 20–50 ensemble runs show probability	3D particles vs depth-averaged; compared in the overlap zone

₹0 data + software licence cost

< 30 min lake detection → evacuation map

Viability: Will it be used?

Fills a real gap

Few tools link simulation, satellite check and HADR plan in one place.

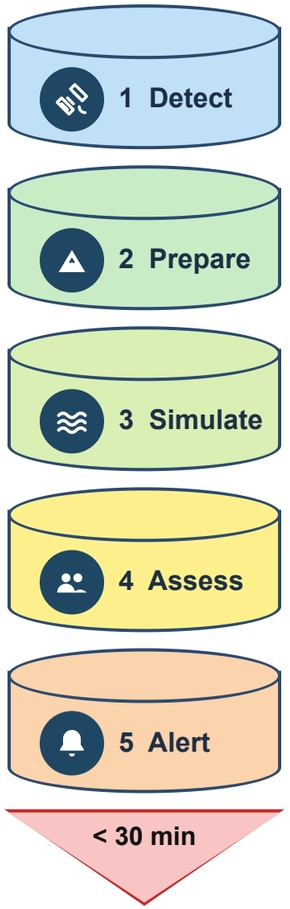
Works where others cannot

Natural dams, landslide lakes and cross-border reservoirs, using satellite data only.

Plugs into India's systems

CAP 1.2 alerts compatible with NDMA SACHET, WMS layers for Bhuvan / QGIS, .shp / .kml for GIS, offline mode.

AUTOMATED FLOW



RISKS → FIXES

⚠ Risk

Heavy solvers are hard to install

Fix: Built-in C# solvers ship in one Docker image; Delft3D is optional

⚠ Risk

No bathymetry, 30 m DEM

Fix: Burn channel into DEM; ensemble runs show uncertainty

⚠ Risk

SPH is compute-heavy

Fix: SPH only near the breach; quick mode skips it; atlas runs are pre-computed

Open open-source core, free data sources

Offline cached-terrain mode for air-gapped sites

Prototype : Click Link

Youtube: Click Link

Github: Click Link